تاریخ: _____

"Solution"

Lectures No.(23-25)(26-28)
Practice Question.

Q1:- Prove the reflexive axioms for (R^2, d) defined by

$$d((x_1,y_1),(x_2,y_2)) = \sqrt{(x_1-x_2)^2+(y_1-y_2)^2} \quad \forall x_1,x_2,y_1,y_2 \in R$$

Solution

m_1 : $(x_1-x_2)^2 \geq 0$

$(y_1-y_2)^2 \geq 0$

$\Rightarrow$ on adding $(x_1-x_2)^2+(y_1-y_2)^2 \geq 0$

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} \geq 0$$

$$d((x_1-x_2),(y_1-y_2)) \geq 0$$

m_1 : $d(x,y) \geq 0$ non-negativity

m_2 : $d(x,y)=0 ; x=y$ Reflexiv

m_3 : $d(x,y)=d(y,x)$ Symme

m_4 : $d(x,y) \leq d(x,z)+d(z,y)$

Triangular inequality

(X,d) → metric space

m_2

$$d((x_1,y_1),(x_2,y_2)) = 0$$

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} = 0$$

$$(x_1-x_2)^2+(y_1-y_2)^2 = 0$$

$$x_1-x_2=0 \; , \; y_1-y_2=0$$

$\therefore |x| = 0$

$x = 0$

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$x_1 = x_2 \quad ; \quad y_1 = y_2$

$(x_1, y_1) = (x_2 = y_2)$

• m_3

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$\because (x-y)^2 = (y-x)^2$

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d((x_1, y_1), (x_2, y_2))$$

• m_4

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$d((x_1, y_1), (x_2, y_2)) \leq \sqrt{(x_1 - x_3 + x_3 - x_2)^2 + (y_1 - y_3 + y_3 - y_2)^2}$$

$$\leq \sqrt{(x_1 - x_3)^2 + (x_3 - x_2)^2 + (y_1 - y_3)^2 + (y_3 - y_2)^2}$$

$$\leq \sqrt{(x_1 - x_3)^2 + (y_1 - y_2)^2 + (x_3 - x_2)^2 + (y_3 - y_2)^2}$$

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$d((x_1,y_1),(x_2,y_2)) \leq d(x_1,y_1)(x_3,y_3) + d(x_3,y_3)(x_2,y_2)$

Q Prove or disprove the reflexive axiom from distance function defined by,

$$d'(x,y) = |x| + |y| \quad , \forall\, x,y \in \mathbb{R}$$

Reflexive property

$[d'(x,y) = 0 \text{ iff } x = y]$

$d'(x,y) = 0$

$|x| + |y| = 0 \qquad \therefore |x| + |y| \Rightarrow |x+y|$

$|x+y| = 0 \qquad \therefore |x| = 0 \Rightarrow x = 0$

$x + y = 0$

$x = -y$

Reflexive property not satisfy

So, d' is not metric on $\mathbb{R}$

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Q Let $P_1(x_1, x_2), P_2(y_1, y_2) \in \mathbb{R}^2$, then prove the non-negativity from the distance function defined by

$$d(P_1, P_2) = \max\{|x_2 - x_1|, |y_2 - y_1|\}$$

Non negativity

$$[\, d(x, y) \geq 0 \,]$$

for give

$$d(P_1, P_2) = \max\{|x_2 - x_1| - |y_2 - y_1|\}$$

$|x_2 - x_1| \geq 0$, $|y_2 - y_1| \geq 0$ $\therefore |x| = 0$

$x = 0$

$$\max\{|x_2 - x_1|, |y_2 - y_1|\} \geq 0$$

$$d(P_1, P_2) \geq 0.$$

Q:- Prove or disprove the Reflexive property from distance function defined by:

$$d(x, y) = |(x - y)(x + y)|$$

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Reflexive property $[d(x,y)=0 \text{ iff } x=y]$

$d(x,y)=0$

$|(x-y)(x+y)|=0$ $\quad \therefore (x-y)(x+y)=x^2-y^2$

$|x^2-y^2|=0$ $\quad \because |x|=0 \Rightarrow x=0$

$x^2-y^2=0$

$x^2=y^2$

$x=\pm y$

$\not\Rightarrow d(x,y)=0 \Rightarrow x=y$

but also $\Rightarrow x=-y$

So the d(distance function is not metric space

Q.. Prove or disprove the Reflexive property of distance function $d': X \times X \to R$ defined as:

$d'(x,y)=\min\{2, d(x,y)\}$, $\forall x,y \in X$

where (X,d) is a metric space.

Reflexive property $[d(x,y)=0 \text{ iff } x=y]$

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$$d'(x,y) = 0$$

$$\Leftrightarrow \quad \min\{2, d(x,y)\} = 0$$

$$\Leftrightarrow d(x,y) = 0 \quad ; \quad 2 \neq 0$$

$$\Leftrightarrow \quad x = y$$

$\therefore$ d is a metric.

d' is reflexive.

Q: Prove or disprove the symmetric property for distance function defined by:

$$d(x,y) = \left| \frac{(x-y)}{xy} \right| \qquad \forall x, y \in \mathbb{R} - \{0\}$$

Symmetric property $[d(x,y) = d(y,x)]$

$$d(x,y) = \left| \frac{(x-y)}{xy} \right|$$

$$= \left| \frac{-(y-x)}{xy} \right|$$

$$= \frac{|(-1)(y-x)|}{|xy|} \Rightarrow = \frac{|y-x|}{xy}$$

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$$= |(y-x)|$$
$$= d(y,x)$$

Q: If d_0 is a discrete metric on $\mathbb{R}$, Then find the value of $d_0(2,2) + d_0(1,2)$

As d_0 is discrete metric on $\mathbb{R}$
Then,

$d_0(2,2) = 0$ (same points)
&
$d_0(1,2) = 1$ (different points)

So $d_0(2,2) + d_0(1,2)$
$= 0 + 1$
$= 1$

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Q. Reduce the expression of distance function

$$d'(x,y) = |x| + |y| = 1, \quad x, y \in \mathbb{R}$$

into system of linear equations

linear equations

$$\pm x \pm y = 1$$

So

$$x + y = 1$$
$$x - y = 1$$
$$-x + y = 1$$
$$-x - y = 1$$

These are linear equations

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تاریخ:

Q. If the distance function is defined by

$$d(x,y) = \sqrt{|x-y|} \quad \forall\ x, y \in \mathbb{R}$$

then solve the equation:

$$d(3x^2, 9x) = 0$$

$$d(x,y) = \sqrt{|x-y|}$$

$$d(3x^2, 9x) = 0$$

So

$$d(3x^2, 9x) = \sqrt{|3x^2 - 9x|} = 0$$

$$= \sqrt{|3|(|x^2 - 3x|)} = 0$$

As $|3| = 3$

$$\Rightarrow \sqrt{3\,|x^2 - 3x|} = 0$$

$$\Rightarrow \sqrt{3}\ \sqrt{|x^2 - 3x|} = 0$$

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تاريخ:

$$\sqrt{|x^2-3x|} = \frac{0}{\sqrt{3}} \quad \geq 0$$

$$\sqrt{|x^2-3x|} \geq 0$$

Squaring on b/s

$$|x^2-3x| \geq 0$$

$$\pm (x^2-3x) = 0$$

$$x^2-3x = 0 \quad , \quad -(x^2-3x) = 0$$

$$x(x-3) = 0 \quad , \quad x(-x+3) = 0$$

$$x = 0 \, , \; x = 3 \quad ; \quad x = 0 \, , \; x = 3$$

So then the Solution will be

$$x = 0 \, , \; x = 3$$

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MTH 405

Practice Questions Solution.

Lect. (29-31)

Q. In the set of all sequence $\{x_n\}$ $n \in N$ such that $\sum_{i=1}^{\infty} |x_i| < \infty$, prove the reflexive property provided that the distance function is given by

$$d(\{x\},\{y\}) = \sum_{k=1}^{\infty} |x_k - y_k|$$

reflexive property $[d(x,y) = 0 \text{ iff } x = y]$

$$d(\{x\},\{y\}) = 0$$

$$\Leftrightarrow \sum_{k=1}^{\infty} |x_k - y_k| = 0.$$

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$\Leftrightarrow |x_1 - y_1| + |x_2 - y_2| + \cdots = 0$ $\quad \because |x| = x \Rightarrow x = 0$

$\Leftrightarrow x_1 - y_1 = 0 \;,\; x_2 - y_2 = 0 \;,\; \cdots$

$\Leftrightarrow x_1 = y_1 \;,\; x_2 = y_2$

$\Leftrightarrow \{x_1, x_2 \cdots\} = \{y_1, y_2 \cdots\}$

$\Leftrightarrow \{x_k\} = \{y_k\}$

Reflexive property satisfy.

Q. If $x_1, x_2, x_3, \ldots, x_n \in \mathbb{R}$, Then by using Cauchy Schwarz Inequality;

Show that

$$(x_1 + x_2 + \cdots + x_n)^2 \leq n(x_1^2 + x_2^2 + \cdots + x_n^2)$$

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Cauchy Schwarz Inequality.

$$\left(\sum_{r=1}^{n}(x_r y_r)\right)^2 \leq \left(\sum_{r=1}^{n} x_r^2\right)\left(\sum_{r=1}^{n} y_r^2\right) \quad -(1)$$

put $y_r = 1$, $\forall\ r = 1, 2, 3 \cdots n$

from eq (1)

$$\Rightarrow \left(\sum_{r=1}^{n}(x_r \cdot 1)\right)^2 \leq \left(\sum_{r=1}^{n} x_r^2\right)\left(\sum_{r=1}^{n} 1^2\right) \quad -(2)$$

$$\therefore \begin{cases} x_r \cdot 1 = x_r & \forall\ r, 1, 2, 3 \cdots n \\ \sum_{r=1}^{n} 1^2 \Rightarrow \sum_{r=1}^{n} 1 = 1 + 1 \cdots 1 (n^{th}\ term) \\ \qquad\qquad = n \end{cases}$$

So, from eq (2)

$$\Rightarrow \left(\sum_{r=1}^{n} x_r\right)^2 \leq n\left(\sum_{r=1}^{n} x_r^2\right)$$

$$(x_1 + x_2 + x_3 + \cdots + x_n)^2 \leq n\left(x_1^2 + x_2^2 + x_3^2 + \cdots + x_n^2\right)$$

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Q. If $x = (x_1, x_2, x_3)$, $y = (y_1, y_2, y_3) \in \mathbb{R}^3$ then prove the non negativity property of the distance function, $d : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}$ defined by $d(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + (x_3-y_3)^2}$

Non Negativity $[d(x,y) \geq 0]$

$x_1 - y_1 \geq 0$, $x_2 - y_2 \geq 0$, $x_3 - y_3 \geq 0$

$(x_1-y_1)^2 \geq 0$, $(x_2-y_2)^2 \geq 0$, $(x_3-y_3)^2 \geq 0$

$(x_1-y_1)^2 + (x_2-y_2)^2 + (x_3-y_3)^2 \geq 0$

$\sqrt{(x_1-y_1)^2 + (x_2-y_2)^2 + (x_3-y_3)^2} \geq 0$

$d(x,y) \geq 0$ Hence proved

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Q: In a metric space (X,d) if $\sup\{d(x,y), \forall x,y \in X\}$ is finite Then show that (X,d) is bounded.

Since

$$\sup\{d(x,y), \forall\ x,y \in X\}$$

is finite.

$\Rightarrow$ diameter of X as a subset of itself i.e. $d(X)$ is finite and therefore X is bounded.

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